Assignment 4: IPFS Integration

Project Overview

This project demonstrates how files can be uploaded and retrieved using IPFS (InterPlanetary File System). It ensures decentralized storage and secure file access using content identifiers (CID).

Objective

To understand how decentralized storage works and how IPFS can be integrated with applications for file handling.

Technologies Used

- IPFS Desktop
- JavaScript / Node.js
- IPFS HTTP Client
- Blockchain (optional integration)

IPFS Integration

Files are uploaded to IPFS using an IPFS client. Each file generates a unique CID (Content Identifier), which is used later to retrieve the file.

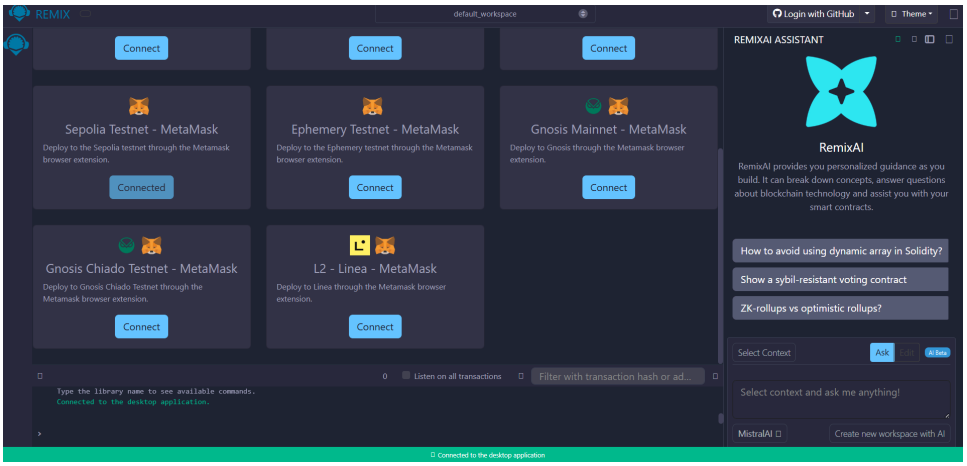
Code (Sample)

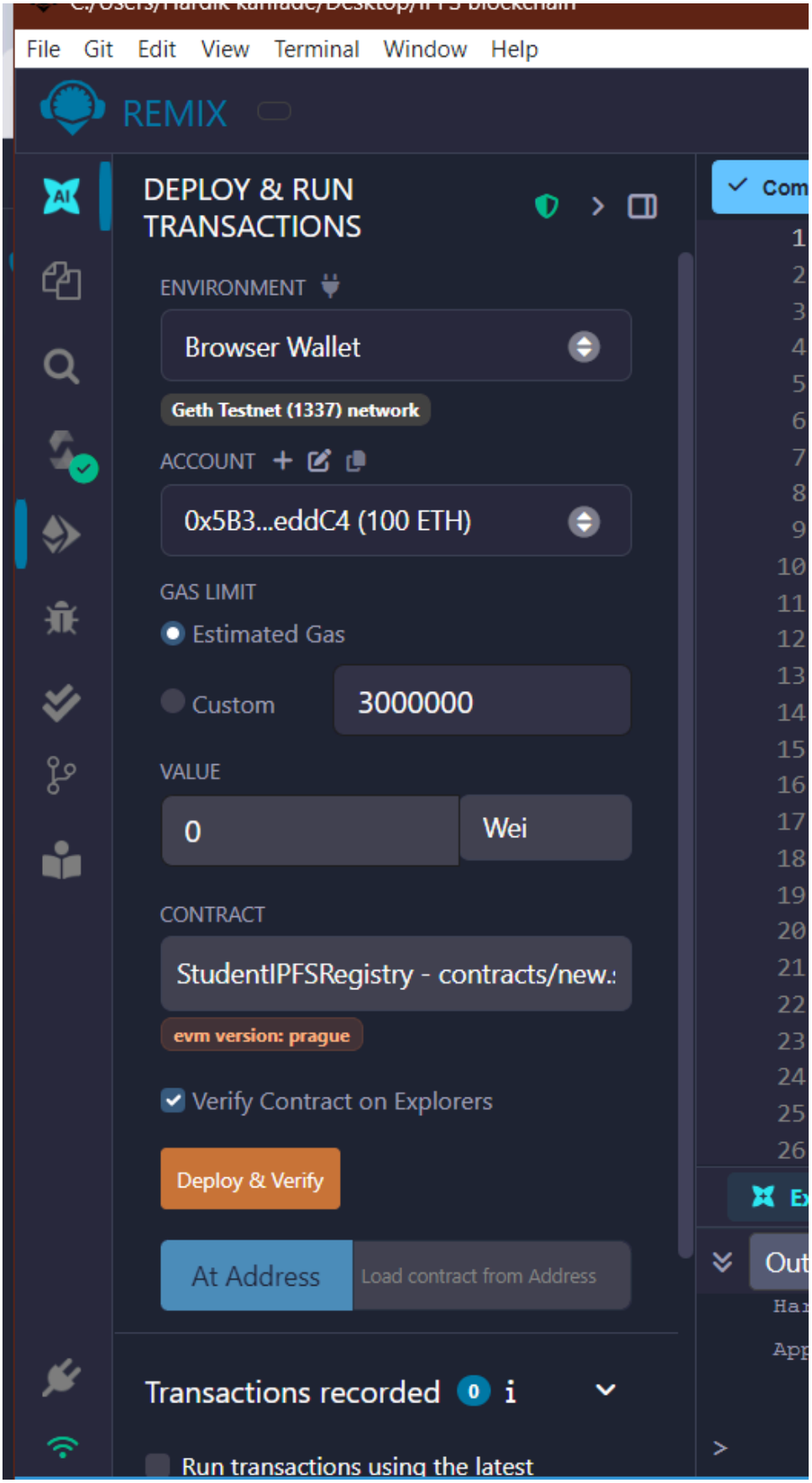
```
// Upload File
const { create } = require('ipfs-http-client');
const ipfs = create('https://ipfs.infura.io:5001/api/v0');

const result = await ipfs.add(file);
console.log(result.path);

// Retrieve File
const data = [];
for await (const chunk of ipfs.cat(result.path)) {
  data.push(chunk);
}
```

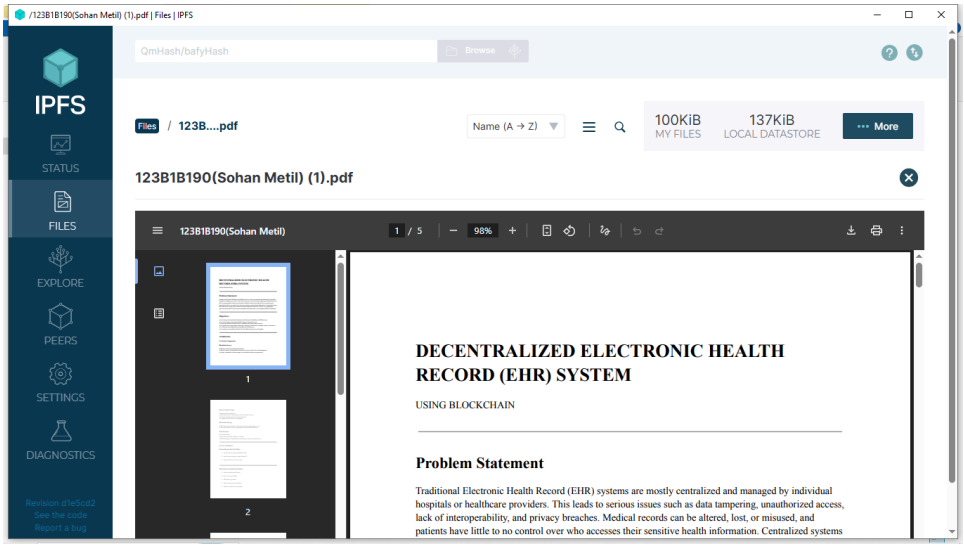
IPFS Hash Example

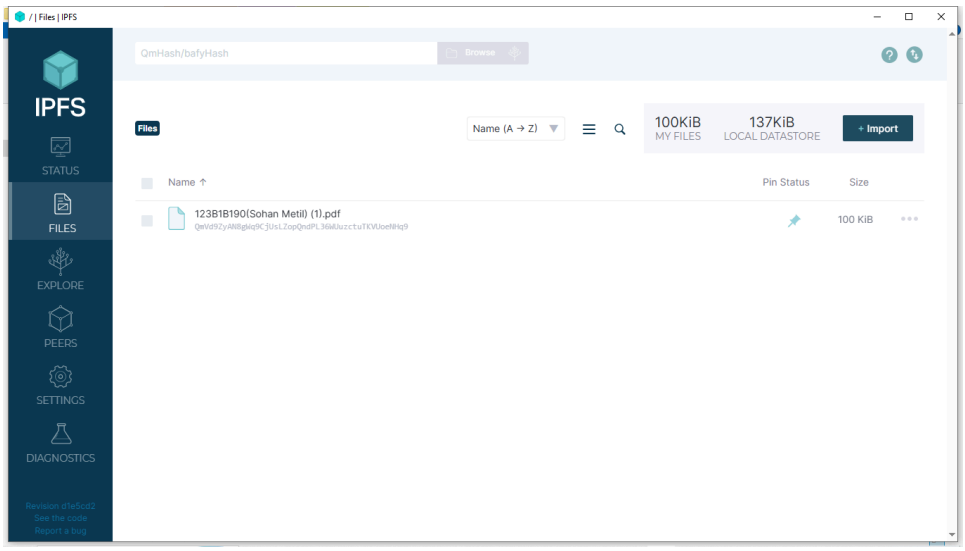
Example CID: QmWxyz123abc456...

Screenshots




File Git Edit View Terminal Window Help
REMIX
DEPLOY & RUN TRANSACTIONS
ENVIRONMENT
Browser Wallet
Geth Testnet (1337) network
ACCOUNT
0x5B3...eddC4 (100 ETH)
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
StudentIPFSRegistry - contracts/new.:
evm version: prague
Verify Contract on Explorers
Deploy & Verify
At Address
Load contract from Address
Transactions recorded 0
Run transactions using the latest





Conclusion

This assignment shows how decentralized storage using IPFS works and how files can be securely stored and retrieved using CIDs.